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UROD uroporphyrinogen decarboxylase [*Homo sapiens* (human)]

Gene ID: 7389, updated on 13-Sep-2026

Summary

Official Symbol	UROD provided by HGNC
Official Full Name	uroporphyrinogen decarboxylase provided by HGNC
Primary source	HGNC:HGNC:12591
See related	Ensembl:ENSG00000126088 MIM:613521; AllianceGenome:HGNC:12591
Gene type	protein coding
RefSeq status	REVIEWED
Organism	Homo sapiens
Lineage	Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo
Also known as	PCT; UPD
Summary	This gene encodes an enzyme in the heme biosynthetic pathway. This enzyme is responsible for catalyzing the conversion of uroporphyrinogen to coproporphyrinogen through the removal of four carboxymethyl side chains. Mutations and deficiency in this enzyme are known to cause familial porphyria cutanea tarda and hepatoerythropoetic porphyria.[provided by RefSeq, Aug 2010]
Expression	Ubiquitous expression in bone marrow (RPKM 73.9), adrenal (RPKM 43.0) and 25 other tissues See more
Orthologs	all

Try the new Gene page
Try the new Transcripts and proteins table

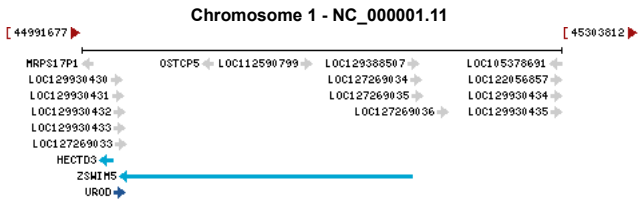
Genomic context

See UROD in Genome Data Viewer

Location: 1p34.1

Exon count: 10

Annotation release	Status	Assembly	Chr	Location
RS_2025_08	current	GRCh38.p14 (GCF_000001405.40)	1	NC_000001.11 (45012254..45015575)
RS_2025_08	current	T2T-CHM13v2.0 (GCF_009914755.1)	1	NC_060925.1 (44882726..44887169)
RS_2024_09	previous assembly	GRCh37.p13 (GCF_000001405.25)	1	NC_000001.10 (45477926..45481247)

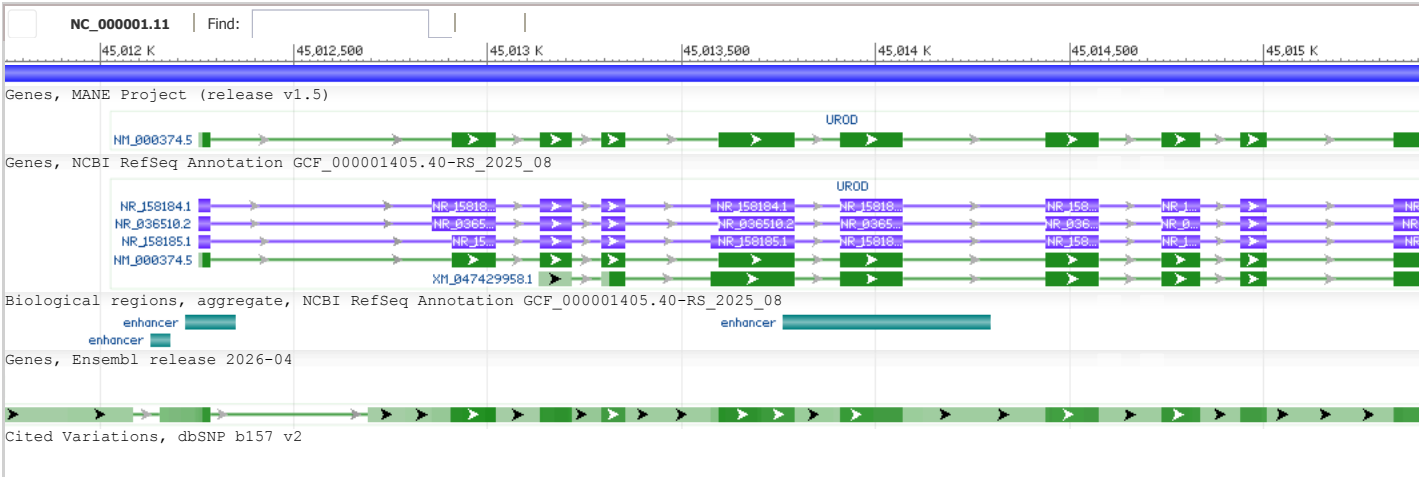


Genomic regions, transcripts, and products

Go to reference sequence details

Genomic Sequence: NC_000001.11 Chromosome 1 Reference GRCh38.p14 Primary Assembly

Go to nucleotide: Graphics FASTA GenBank



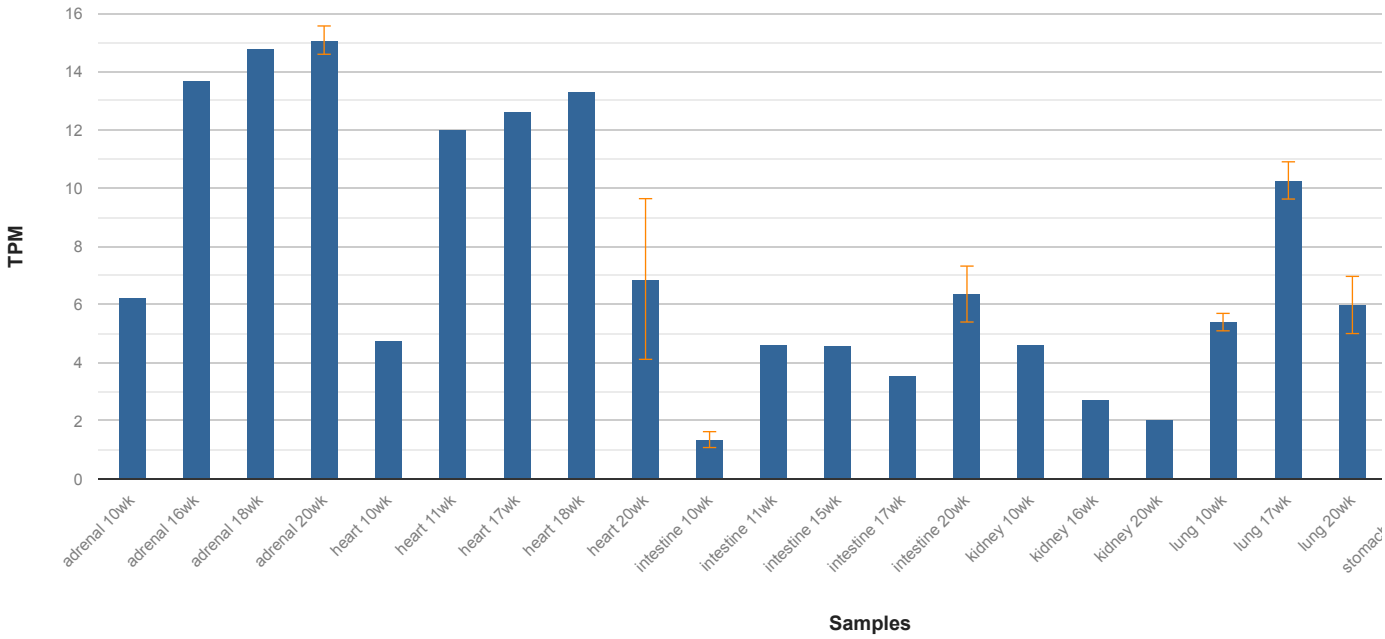


Expression

See details

Tissue-specific circular RNA induction during human fetal development

- Project title: Tissue-specific circular RNA induction during human fetal development
- Description: 35 human fetal samples from 6 tissues (3 - 7 replicates per tissue) collected between 10 and 20 weeks gestational time were sequenced using Illumina TruSeq Stranded Total RNA
- BioProject: PRJNA270632
- Publication: PMID 26076956
- Analysis date: Mon Apr 2 22:54:59 2018



Bibliography

Related articles in PubMed

1. Molecular analysis of the UROD gene in 17 Argentinean patients with familial porphyria cutanea tarda: characterization of four novel mutations.
   Méndez M, *et al*. Mol Genet Metab, 2012 Apr. PMID 22382040
2. Hepatoerythropoietic porphyria due to a novel mutation in the uroporphyrinogen decarboxylase gene.
   To-Figueras J, *et al*. Br J Dermatol, 2011 Sep. PMID 21668429, **Free PMC Article**
3. Three new mutations in the uroporphyrinogen decarboxylase gene in familial porphyria cutanea tarda. Mutation in brief no. 237. Online.
   McManus JF, *et al*. Hum Mutat, 1999. PMID 10338097
4. Screening for mutations in the uroporphyrinogen decarboxylase gene using denaturing gradient gel electrophoresis. Identification and characterization of six novel mutations associated with familial PCT.
   Christiansen L, *et al*. Hum Mutat, 1999. PMID 10477430
5. Molecular heterogeneity of familial porphyria cutanea tarda in Spain: characterization of 10 novel mutations in the UROD gene.
   Méndez M, *et al*. Br J Dermatol, 2007 Sep. PMID 17627795

See all (84) citations in PubMed

GeneRIFs: Gene References Into Functions

What's a GeneRIF?

1. These results expand the molecular heterogeneity of Porphyria Cutanea Tarda and Hepatoerythropoietic Porphyria by adding a total of 19 novel UROD mutations.
2. a new mutation in the UROD gene in Egyptian patients with Hepatoerythropoietic porphyria, is reported.
3. a new insight in the conformational changes occurred in the mutant structures of UROD protein.
4. Data suggest that the traditional Chinese medicine (TCM) candidate potential three-in-one inhibitors for three drug target proteins epidermal growth factor receptor (EGFR), Her2, and uroporphyrinogen decarboxylase (UROD) against head and neck cancer.
5. Expression of the D306Y mutation results in an insoluble recombinant protein. G318R and K297N have little effect on the structure or activity of recombinant URO-D, but the proteins display reduced stability in vitro.
6. Among seventeen F-PCT patients, sixteen UROD mutations were identified.
7. analysis of uroporphyrinogen decarboxylase as a potential target for specific components of traditional Chinese medicine
8. Mutations in UROD gene is associated with familial porphyria cutanea tarda.
9. Hepatoerythropoietic porphyria and familial porphyria cutanea tarda G281E mutation in the uroporphyrinogen decarboxylase gene
10. function of Arg37 in uroporphyrinogen III decarboxylase

Submit: New GeneRIF Correction See all GeneRIFs (17)

Phenotypes

BioGRID CRISPR Screen Phenotypes (605 hits/1394 screens)

Find tests for this gene in the NIH Genetic Testing Registry (GTR)

Review eQTL and phenotype association data in this region using PheGenI

Associated conditions

Description	Tests
Familial porphyria cutanea tarda MedGen: C0268323, OMIM: 176100, GeneReviews: Familial Porphyria Cutanea Tarda, Hepatoerythropoietic Porphyria	Compare labs
Hepatoerythropoietic porphyria MedGen: C0162569, GeneReviews: Hepatoerythropoietic Porphyria	Compare labs

Copy number response

Description
Copy number response Triplosensitivity No evidence available (Last evaluated 2024-10-04) ClinGen Genome Curation Page
Haploinsufficency Some evidence for dosage pathogenicity (Last evaluated 2024-10-04) ClinGen Genome Curation Page, PubMed

Variation

See variants in ClinVar

See studies and variants in dbVar

See Variation Viewer (GRCh37.p13)

HIV-1 interactions

Protein interactions

Protein	Gene	Interaction	Pubs
Vpu	vpu	HIV-1 Vpu-expressing U937 monocytes coculture with LX2 stellate cells to upregulate expression of profibrogenic markers COL-1, PCT, SMA-1, VEGF, and MMP2, which is inhibited by MIF treatment	PubMed

Go to the HIV-1, Human Interaction Database

Pathways from PubChem

16 items Download

Porphyrin_metabolism (Porphyrin metabolism)
Data Source: INOH External ID: MI0036639
Taxonomy Name: Homo sapiens (human)

Porphyrin metabolism
Data Source: Reactome External ID: R-HSA-189445
Taxonomy Name: Homo sapiens (human)

Porphyrin Metabolism
Data Source: PathBank External ID: SMP0000024
Taxonomy Name: Homo sapiens (human)

heme biosynthesis
Data Source: BioCyc External ID: PWY-5920
Taxonomy Name: Homo sapiens (human)

Heme biosynthesis
Data Source: WikiPathways External ID: WP561
Taxonomy Name: Homo sapiens (human)

Congenital Erythropoietic Porphyria (CEP) or Gunther Disease
Data Source: PathBank External ID: SMP0000345
Taxonomy Name: Homo sapiens (human)

heme biosynthesis from uroporphyrinogen-III I
Data Source: BioCyc External ID: HEME-BIOSYNTHESIS-II
Taxonomy Name: Homo sapiens (human)

Porphyria Variegata (PV)
Data Source: PathBank External ID: SMP0000346
Taxonomy Name: Homo sapiens (human)

Hereditary Coproporphyria (HCP)
Data Source: PathBank External ID: SMP0000342
Taxonomy Name: Homo sapiens (human)

Hemesynthesis defects and porphyrias
Data Source: WikiPathways External ID: WP5169
Taxonomy Name: Homo sapiens (human)

Metabolism
Data Source: Reactome External ID: R-HSA-1430728
Taxonomy Name: Homo sapiens (human)

Heme biosynthesis
Data Source: Reactome External ID: R-HSA-189451
Taxonomy Name: Homo sapiens (human)

Acute Intermittent Porphyria
Data Source: PathBank External ID: SMP0000344
Taxonomy Name: Homo sapiens (human)

HMOX1 pathway (COVID-19 Disease Map)
Data Source: COVID-19 Disease Map External ID: 718
Taxonomy Name: Homo sapiens (human)

heme b biosynthesis I (aerobic)
Data Source: BioCyc External ID: HEME-BIOSYNTHESIS-II

superpathway of b heme biosynthesis from glycine
Data Source: BioCyc External ID: PWY-5920

SIDEBAR

PubChem

Interactions

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Products	Interactant	Other Gene	Complex	Source	Pubs	Description
P06132	P06132	UROD		HPRD	PubMed	
BioGRID:113235	BioGRID:121864	ACE2		BioGRID	PubMed	Affinity Capture-MS
BioGRID:113235	BioGRID:106825	APEX1		BioGRID	PubMed	Affinity Capture-RNA
BioGRID:113235	BioGRID:123745	ATG10		BioGRID	PubMed	Phenotypic Suppression
BioGRID:113235	BioGRID:122171	ATG3		BioGRID	PubMed	Phenotypic Suppression
BioGRID:113235	BioGRID:115787	ATG7		BioGRID	PubMed	Phenotypic Suppression
BioGRID:113235	BioGRID:114907	BAG3		BioGRID	PubMed	Affinity Capture-Luminescence; Affinity Capture-MS
BioGRID:113235	BioGRID:107532	CHD3		BioGRID	PubMed	Two-hybrid
BioGRID:113235	BioGRID:119904	CHRAC1		BioGRID	PubMed	Co-fractionation
BioGRID:113235	BioGRID:121724	CPNE5		BioGRID	PubMed	Affinity Capture-MS
BioGRID:113235	BioGRID:4263559	E2		BioGRID	PubMed	Proximity Label-MS
BioGRID:113235	BioGRID:971751	EBNA-1		BioGRID	PubMed	Affinity Capture-MS
BioGRID:113235	BioGRID:108276	EGFR		BioGRID	PubMed	Negative Genetic; PCA
BioGRID:113235	BioGRID:116298	FAF1		BioGRID	PubMed	Two-hybrid
BioGRID:113235	BioGRID:117326	FBXO7		BioGRID	PubMed	Biochemical Activity
BioGRID:113235	BioGRID:108887	GARS1		BioGRID	PubMed	Affinity Capture-MS
BioGRID:113235	BioGRID:115353	HMGXB4		BioGRID	PubMed	Two-hybrid
BioGRID:113235	BioGRID:110043	KRAS		BioGRID	PubMed	Negative Genetic
BioGRID:113235	BioGRID:127609	LRATD2		BioGRID	PubMed	Two-hybrid
BioGRID:113235	BioGRID:110484	MOV10		BioGRID	PubMed	Affinity Capture-RNA
BioGRID:113235	BioGRID:110611	MTAP		BioGRID	PubMed	Co-fractionation
BioGRID:113235	BioGRID:117673	MYCBP		BioGRID	PubMed	Co-fractionation
BioGRID:113235	BioGRID:4383847	N		BioGRID	PubMed	Proximity Label-MS
BioGRID:113235	BioGRID:110894	NME1		BioGRID	PubMed	Co-fractionation
BioGRID:113235	BioGRID:115745	NXF1		BioGRID	PubMed	Affinity Capture-RNA

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General gene information

Markers

Homology

NCBI Orthologs
Orthologs from OrthoDB

Gene Ontology Provided by GOA

Function	Evidence Code	Pubs
enables protein binding	IPI	PubMed
enables uroporphyrinogen decarboxylase activity	IBA	
enables uroporphyrinogen decarboxylase activity	IDA	PubMed
enables uroporphyrinogen decarboxylase activity	IEA	
enables uroporphyrinogen decarboxylase activity	TAS	PubMed

Process	Evidence Code	Pubs
involved_in heme A biosynthetic process	IEA	
involved_in heme B biosynthetic process	IDA	PubMed
involved_in heme B biosynthetic process	IEA	
involved_in heme biosynthetic process	IBA	
involved_in heme biosynthetic process	IC	PubMed
involved_in porphyrin-containing compound biosynthetic process	IEA	
involved_in porphyrin-containing compound catabolic process	IDA	PubMed
involved_in porphyrin-containing compound metabolic process	IDA	PubMed

Component	Evidence Code	Pubs
is_active_in cytosol	IBA	
located_in cytosol	IDA	
located_in cytosol	IEA	
is_active_in cytosol	ISS	
located_in cytosol	TAS	
located_in nucleoplasm	IDA	

SIDEBAR

General protein information

Preferred Names
uroporphyrinogen decarboxylase

Names
uroporphyrinogen III decarboxylase

NP_000365.3
EC 4.1.1.37

XP_047285914.1
EC 4.1.1.37

XP_054194585.1
EC 4.1.1.37

XP_054194586.1
EC 4.1.1.37

NCBI Reference Sequences (RefSeq)

NEW **Try the new Transcript table**

RefSeqs maintained independently of Annotated Genomes

These reference sequences exist independently of genome builds. Explain

Genomic

1. **NG_007122.2 RefSeqGene**

Range	5097..8418
Download	GenBank, FASTA, Sequence Viewer (Graphics), LRG_1079

mRNA and Protein(s)

1. NM_000374.5 → NP_000365.3 **uroporphyrinogen decarboxylase**

   See identical proteins and their annotated locations for NP_000365.3

   Status: REVIEWED

Description	Transcript Variant: This variant (1) represents the shortest transcript and encodes the functional protein.
Source sequence(s)	AK291877, AL359473
Consensus CDS	CCDS518.1
UniProtKB/Swiss-Prot	A8K762, P06132, Q16863, Q16883, Q53YB8, Q53ZP6, Q6IB28, Q9BUZ0
UniProtKB/TrEMBL	A0ACI8S216
Related	ENSP00000246337.4, ENST00000246337.9

Conserved Domains (1) summary

	cd00717 Location:20 → 359	URO-D; Uroporphyrinogen decarboxylase (URO-D) is a dimeric cytosolic enzyme that decarboxylates the four acetate side chains of uroporphyrinogen III (uro-III) to create coproporphyrinogen III, without requiring any prosthetic groups or cofactors. This reaction ...

RNA

1. NR_036510.2 **RNA Sequence**

   Status: REVIEWED

Description	Transcript Variant: This variant (2) uses an alternate splice site, compared to variant 1. This variant is represented as non-coding because the use of the 5'-most supported translational start codon, as used in variant 1, renders the transcript a candidate for nonsense-mediated mRNA decay (NMD).
Source sequence(s)	AK225053, AK291877, AL359473
Related	ENST00000651476.1

2. NR_158184.1 **RNA Sequence**

   Status: REVIEWED

Source sequence(s)	AL359473

3. NR_158185.1 **RNA Sequence**

   Status: REVIEWED

Source sequence(s)	AL359473
Related	ENST00001033697.1

RefSeqs of Annotated Genomes: GCF_000001405.40-RS_2025_08

The following sections contain reference sequences that belong to a specific genome build. Explain

Reference GRCh38.p14 Primary Assembly

Genomic

1. **NC_000001.11 Reference GRCh38.p14 Primary Assembly**

Range	45012254..45015575
Download	GenBank, FASTA, Sequence Viewer (Graphics)

mRNA and Protein(s)

1. **XM_047429958.1 → XP_047285914.1 uroporphyrinogen decarboxylase isoform X1**

UniProtKB/TrEMBL	A0ACI8TWL9
Related	ENSP00000607064.1, ENST00000937005.2

Alternate T2T-CHM13v2.0

Genomic

1. **NC_060925.1 Alternate T2T-CHM13v2.0**

Range	44882726..44887169
Download	GenBank, FASTA, Sequence Viewer (Graphics)

mRNA and Protein(s)

1. **XM_054338610.1 → XP_054194585.1 uroporphyrinogen decarboxylase isoform X2**

UniProtKB/TrEMBL	A0A494C0Q8, A0ACI8S216

2. **XM_054338611.1 → XP_054194586.1 uroporphyrinogen decarboxylase isoform X1**

UniProtKB/TrEMBL	A0ACI8TWL9

Related sequences

Items 1 - 25 of 46 < Prev Page 1 of 2 Next >

Nucleotide		Protein
Heading	Accession and Version	
genomic	AF047383.1	AAC03563.1
genomic	AH007229.2	AAD04167.1
		AAD04168.1
genomic	AL359473.22 (17955..21491)	None
genomic	AY292986.1	AAP44118.1
genomic	CH471059.2	EAX07007.1
		EAX07008.1
genomic	CP068277.2 (44882726..44887169)	None
genomic	M60891.1	AAB59456.1
genomic	U30787.1	AAC50482.1
genomic	X89267.1	CAA61540.1
mRNA	AF104421.1	AAD04571.1
mRNA	AF104422.1	AAD04572.1
mRNA	AF104423.1	AAD04573.1
mRNA	AF104424.1	AAD04574.1
mRNA	AF104425.1	AAD04575.1
mRNA	AF104426.1	AAD04576.1
mRNA	AF104427.1	AAD04577.1
mRNA	AF104428.1	AAD04578.1
mRNA	AF104429.1	AAD04579.1
mRNA	AF104430.1	AAD04580.1
mRNA	AF104431.1	AAD04581.1
mRNA	AF104432.1	AAD04582.1
mRNA	AF104433.1	AAD04583.1
mRNA	AF104434.1	AAD04584.1

Nucleotide		Protein
Heading	Accession and Version	
mRNA	AF104435.1	AAD04585.1
mRNA	AF104436.1	AAD04586.1

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Protein Accession	Links	
	GenPept Link	UniProtKB Link
P06132.2	GenPept	UniProtKB/Swiss-Prot:P06132

SIDEBAR

Additional links

Locus-specific Databases

UROD database

Gene LinkOut